ASSIL HAMDI MEMORIAL MATHEMATICS OLYMPIAD

Official Examination Paper - 4 Hours Strict Limit

Rules: 1. Legible handwriting required. 2. Fully justified proofs mandatory.
3. Zero tolerance for AI or plagiarism: Results in score invalidation and account ban.

Problem 1 (Geometry - 7 Points):

Let ABC be an acute triangle with circumcircle Gamma. Let D be the foot of the altitude from A to BC, and M be the midpoint of BC. The line AD meets Gamma again at X. Let P be the intersection of AM with Gamma (P != A). The tangent to Gamma at P meets BC at T. The line TD meets Gamma again at K. Prove that A, M, K, D are concyclic.

Problem 2 (Number Theory - 7 Points):

Find all pairs of positive integers (n, p) with p prime such that:

n^p + p^n = (n + p)!

Problem 3 (Algebra - 7 Points):

Find all functions f : R -> R such that for all real numbers x, y:

f(x * f(x + y)) = f(y * f(x)) + x^2

Problem 4 (Combinatorics - 7 Points):

On an n x n board (n >= 3 odd), n^2 - 1 checkers are placed leaving the central cell empty. A valid jump moves a checker over an adjacent one into an empty space, removing it. Determine if it is possible to reduce the board to exactly one remaining checker.